

R3: Noise-Floor Mis-specification

How accurately must the sensor noise PSD be known?

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SHARC buoy project Code: `runR3NoiseMisspecification.m`, commit 0389536 Results: `Results/R3/R3_20261008_190330`

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 2 Oct run were low by $\alpha(\omega) = H\omega^2/g$; `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08). Settings unchanged. This summary supersedes R3_20261002.124033. The reference-channel SNR rose about fourfold (LSM6DSV16X: minimum 18 to 70), and with it the conclusion changed.

Summary

Question. The transmissibility estimator subtracts an assumed reference-noise PSD. If that assumption is wrong by a fraction ϵ , how biased are β and R ? **Outcome.** At the corrected signal level the noise floor is **no longer a critical calibration**. For the LSM6DSV16X even omitting the correction altogether gives only $d_{\text{sys}} = 0.129$ and $\delta\beta/\beta < 0.001\%$; no tolerance ($|\delta\beta/\beta| \leq 1\%$, $d_{\text{sys}} \leq 0.5$ or 1) is reached anywhere in $[-100\%, +50\%]$. The same holds for every sensor except the noisiest, the LSM6DS3TR-C, which needs the floor to within $\pm 65\%$ for $d_{\text{sys}} \leq 0.5$ (no correction: $d_{\text{sys}} = 0.76$). The measured colour of the noise is irrelevant ($d_{\text{sys}} \leq 0.007$). **Controlling quantity:** the reference-channel SNR.

1 What the study does

Why only the reference noise matters. The estimator is

$$\hat{T}_j = \frac{\hat{S}_{jr}}{\hat{S}_{rr} - S_{nn,\text{assumed}}}.$$

Noise on a non-reference sensor is uncorrelated with the reference, so it averages out of $\hat{S}_{jr}$ and adds variance but no bias. Noise on the reference enters $\hat{S}_{rr}$ in expectation and must be subtracted. An error in that subtraction biases every transmissibility by the same factor.

The bias. With $S_{nn,\text{assumed}} = (1 + \epsilon)S_{nn}$, the error is $\Delta = \epsilon S_{nn}$ ($\epsilon = -100\%$ is no correction). Using the window-smoothed spectra, the additional feature bias is

$$b_{\text{noise}} = T_W \frac{\Delta}{S_{rr,\text{sig},W} - \Delta} \approx T_W \epsilon \frac{S_{nn}}{S_{rr,\text{sig}}} \quad (\text{small } \epsilon),$$

exact in expectation for a fixed bin set; the sensitivity is set by the noise-to-signal ratio of the reference. Over-estimating the floor ($\epsilon > 0$) shrinks the denominator, so positive errors grow faster.

Projection. As in R4, $\delta\theta = F^{-1}J^T\Sigma^{-1}b_{\text{noise}}$, $d_{\text{sys}} = \sqrt{\delta\theta^T F \delta\theta}$, $\delta\beta/\beta = e^{\delta \ln \beta} - 1$, and for the class the bias adds as a vector to the Welch bias. Setting: p_0 , Level 2C, $N = 65536$, $L = 2048$, JONSWAP $H_s = 0.05$ m, all six noise cases; pre-registered $\epsilon \in \{-100, -50, -30, -10, +10, +30, +50\}\%$; tolerances are first crossings on a 1501-point grid over $[-100\%, +50\%]$. Runtime 18 s, no EMM solves.

Checks. The bias passed through `evaluateScenario`'s own systematic-bias path gives the same d_{sys} (0.038574 at $\epsilon = +30\%$, LSM6DSV16X; relative difference 0). The coloured cases use the *measured* basin-band noise shape (power law fitted to the 16.9 h static record) while the estimator subtracts its flat band mean; fitted band means match the configured S_{nn} to within 1%.

2 Results

Required noise-floor accuracy (smaller $|\epsilon|$ of the two signs; search range $[-100\%, +50\%]$).

Noise case	S_{nn}	min SNR $S_{rr,\text{sig}}/S_{nn}$	tolerance $ \epsilon $			no correction ($\epsilon = -100\%$)	
			$ \delta\beta/\beta \leq 1\%$	$d_{\text{sys}} \leq 0.5$	$d_{\text{sys}} \leq 1$	d_{sys}	$\delta\beta/\beta$
Datasheet grade	6.5×10^{-8}	861	n.r.	n.r.	n.r.	0.025	0.000%
Three IMUs fused	5.0×10^{-7}	112	n.r.	n.r.	n.r.	0.093	0.000%
LSM6DSV16X	8.0×10^{-7}	70	n.r.	n.r.	n.r.	0.129	0.000%
G3 assumption	1.42×10^{-6}	40	n.r.	n.r.	n.r.	0.191	0.000%
ICM-42688-P	1.52×10^{-6}	37	n.r.	n.r.	n.r.	0.201	0.000%
LSM6DS3TR-C	1.24×10^{-5}	5	n.r.	65.4%	n.r.	0.763	-0.008%

n.r.: not reached within $[-100\%, +50\%]$. min SNR: smallest reference signal-to-noise ratio over the accepted bins. 0.000%: below $5 \times 10^{-4}\%$ in magnitude. Pre-amendment (2 Oct), LSM6DSV16X: min SNR 18, tolerance $\pm 30.6\%$ for $d_{\text{sys}} \leq 0.5$, no correction $d_{\text{sys}} = 1.63$.

Coloured noise (true = measured shape; assumed = flat band mean).

Sensor	Fitted PSD shape	Max $ \Delta /S_{nn}$	$\delta\beta/\beta$	d_{sys}	d_{tot} (vs d_W alone)
LSM6DSV16X	$\omega^{-0.15}$	0.09	+0.0000%	0.0015	0.436 (0.436)
ICM-42688-P	$\omega^{-0.06}$	0.03	+0.0000%	0.0010	0.387 (0.387)
LSM6DS3TR-C (pre-registered)	$\omega^{-0.10}$	0.05	+0.0002%	0.0071	0.236 (0.237)

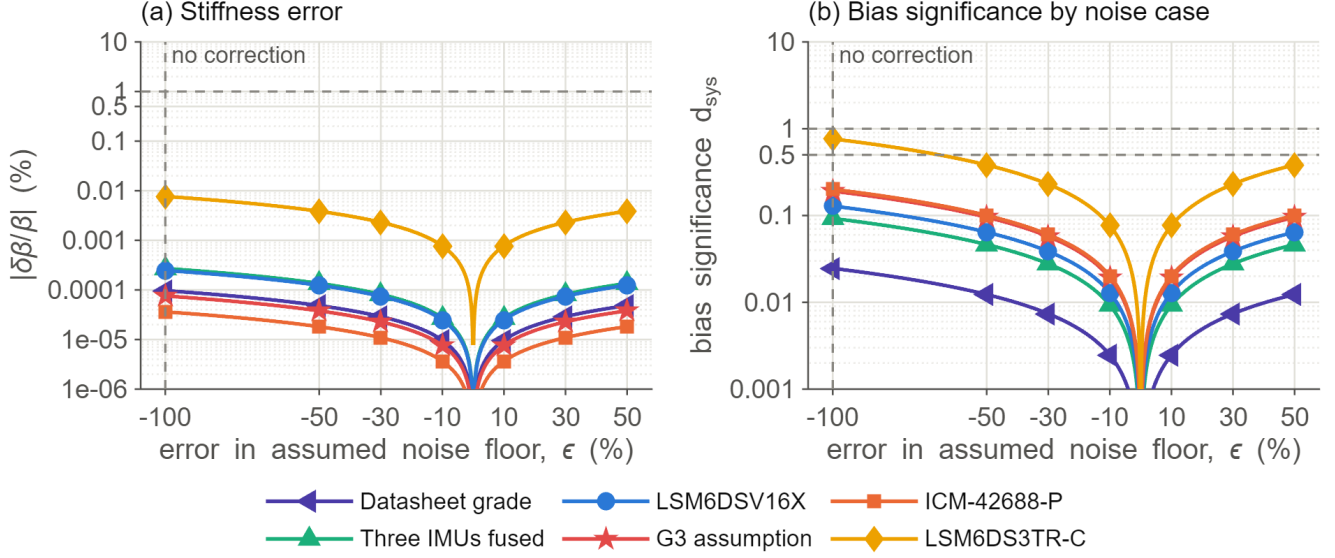


Figure 1. (a) Stiffness error $|\delta\beta/\beta|$ and (b) bias significance d_{sys} against the error ϵ in the assumed reference noise floor, for every noise case (log scale; corrected FRF). Curves: fine grid; markers: pre-registered ϵ . Ticks at 1% in (a) and at $d = 0.5, 1$ in (b). Dashed line: no correction ($\epsilon = -100\%$).

3 Interpretation

- The correction is now optional for the candidate sensors.** With the reference SNR at 70 or more, the subtraction changes d_{sys} by at most 0.13 for the LSM6DSV16X and 0.20 for the ICM-42688-P, and the stiffness by less than 0.001%. It remains good practice (it costs nothing and protects calmer seas), but it is not a calibration requirement at the reference sea state.
- What changed.** The noise bias scales as $S_{nn}/S_{rr,sig}$. The corrected signal level raised the reference SNR about fourfold, and the bias fell faster than the uncertainty: d_{sys} without correction dropped from 1.63 to 0.129 for the LSM6DSV16X. This is the opposite trend to D2 and R1, where the Welch bias, which does not depend on the noise, became more significant as σ fell.
- A simple rule still holds.** The bias is nearly linear in ϵ , so $|\epsilon|_{d=0.5} \approx 0.5/d_{sys}(\text{no correction})$: LSM6DS3TR-C $0.5/0.763 = 65.5\%$ against 65.4% found. One evaluation without correction gives the tolerance for any sensor, floe or sea; a tolerance exists only where $d_{sys}(\text{no correction}) > 0.5$.
- Spectral colour does not matter.** The measured spectra are nearly flat over the band ($\omega^{-0.06}$ to $\omega^{-0.15}$); replacing them by their band mean adds at most $d_{sys} = 0.007$ and leaves d_{tot} at the Welch value. A frequency-dependent noise model is not needed.

Limitations.

- SNR depends on the forcing.** The noise bias scales as $S_{nn}/S_{rr,sig}$, so in calmer seas than $H_s = 0.05$ m the correction matters again: halving H_s quarters the SNR and quadruples the bias per unit ϵ (the tolerance tightens by less, because σ grows too). For the noisiest sensor, or small waves, use the rule above.
- Lab versus deployment noise.** S_{nn} was measured in a static lab test; mounting, temperature and platform vibration can change it in the field.
- Fixed bin set and covariance.** Bins accepted under the true noise are kept; Σ_k uses the true white level; the coloured case is a bias test only.
- One floe.** p_0 , Level 2C, one sea state.

Requirement for the experiment and the thesis

Keep the reference-noise correction with the flat band-mean S_{nn} of the reference sensor, but it is not a critical calibration: for the LSM6DSV16X or ICM-42688-P at the reference sea state, even no correction keeps $d_{sys} \leq 0.2$. A tolerance applies only to low-SNR conditions (noisy sensors or calm seas); it is $0.5/d_{sys}(\text{no correction})$, $\pm 65\%$ for the LSM6DS3TR-C.

Outputs (Results/R3/): R3.20261008.190330.log/.mat/_summary.txt; R3.sweep.csv; R3.tolerance.csv; R3.coloured.csv; figures/R3_fig1_noise_misspecification (.pdf/.png). Superseded: R3.20261002.124033 (pre-amendment).